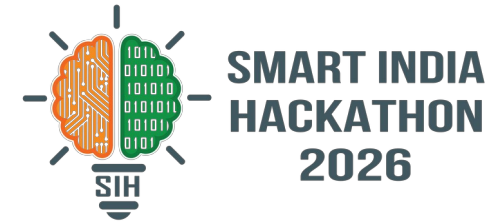


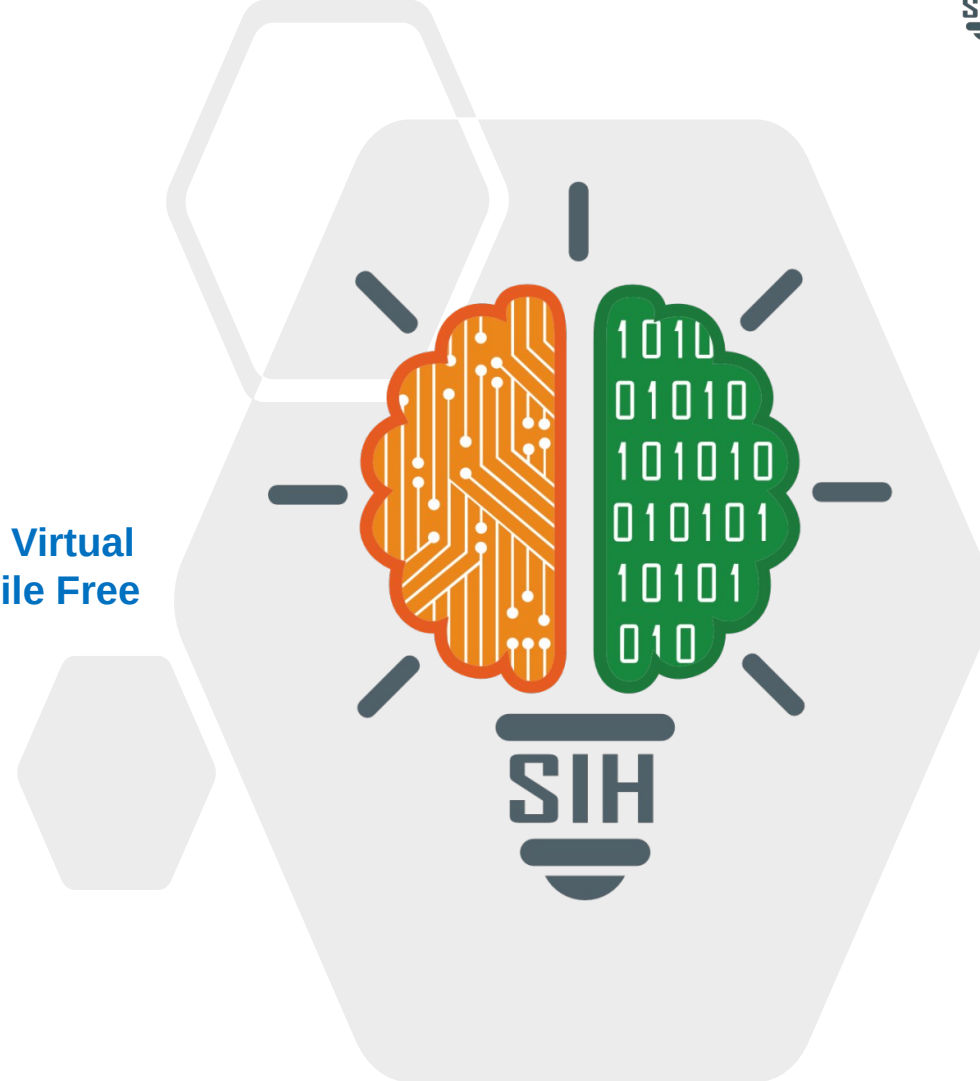
SMART INDIA HACKATHON 2026



ANVESH

Belief-Driven Coarse PAT for Mobile FSOC Terminals

- Problem Statement ID : [SIH26169](#)
- Problem Statement Title : [Development of an AI-Based Virtual Camera Tracking System for Coarse Alignment of Mobile Free Space Optical Communication \(FSOC\) Terminals](#)
- Theme : [Smart Automation / Space Technology](#)
- PS Category : [Software](#)
- Team ID : _____
- Team Name (Registered on portal) : [Regnum Carya](#)



Where to look · how wide to look · what not to chase

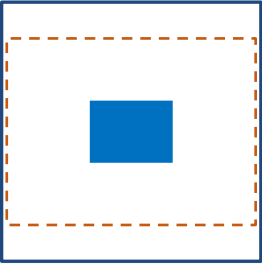
IDEA TITLE

❖ Proposed Solution (Describe your Idea/Solution/Prototype)

ANVESHA: a probability map of the beacon decides **where to point** and **how wide to look**.

Measured (SIL, 48 runs each): 87 % of runs acquired ≤ 2 s
vs 12–15 % for spiral baselines (slide 4)

How it addresses the problem



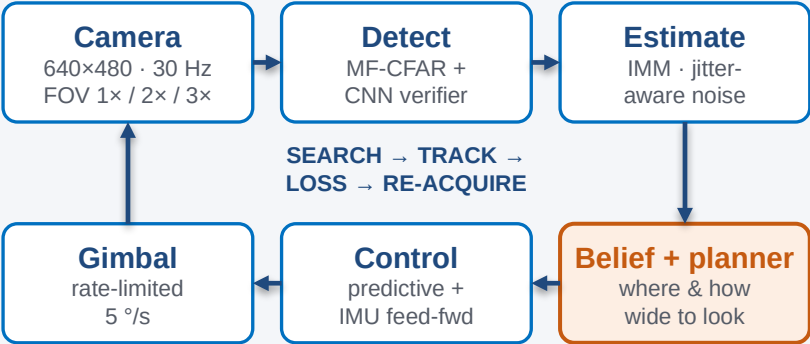
4°×3° FOV = **7.7 %** of the screen
5 °/s gimbal $\Rightarrow \leq 10^\circ$ slew in 2 s
Blind raster needs $\approx 40\text{--}50^\circ$ of slew
 $\Rightarrow$ **~8–10 s**
Measured spiral baselines A / B:
median **7.8 / 9.2 s**
PS asks ≤ 2 s from a random start

To scale: screen 12.5°×12.5° · ■ 4°×3° base FOV · ▭ 12°×9° wide FOV

Detecting a bright spot is easy. Knowing **where to look** is the real problem.

Detailed explanation — the closed loop

▼ disturbances in: noise · jitter · fog/haze/rain · platform motion
▼ or evaluator .mp4 video (Benchmark-2): PTZ bypassed



Out: per-frame truth-based metrics $\rightarrow$ CSV · JSON · HTML report (auto)

Innovation and uniqueness

1

Belief-driven search
every empty frame is evidence: probability is removed where the beacon was not seen; next look = best P(detect) per second

2

Disturbance-aware FOV
FOV (1×/2×/3×) chosen by predicted detection probability from the measured noise: go wide only when the beacon stays detectable

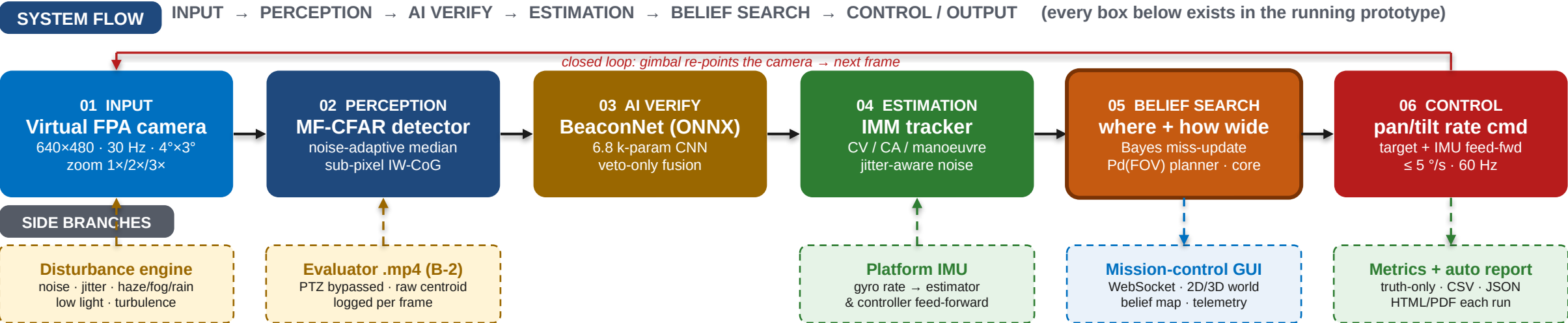
3

One belief, whole lifecycle
on loss the search restarts from the tracker's prediction; jitter-aware estimator stops the gimbal chasing jitter

Typical public approach	Limitation	ANVESHA approach
Fixed spiral / raster at base FOV	Open loop; ignores empty frames; ~8–10 s worst case	Recursive Bayesian belief + next-best-look planner
Wide-FOV mode switched by a fixed rule	SNR loss of a wider FOV is not modelled	FOV picked from a measured-noise detection model
Re-acquire = restart spiral / Kalman coast	Throws away what the tracker knew	Belief re-seeded from IMM predicted mean & covariance

Based on our review of 134 papers and 20 public SIH26169 repositories, we did not identify a coarse-PAT system that combines these three elements.

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)



❖ Analysis of the feasibility of the idea

Measured in simulation (software-in-the-loop), 3 seeds · 16 scenarios · same seeds for all pipelines · baseline PID gains grid-tuned · laptop CPU only (no GPU used)

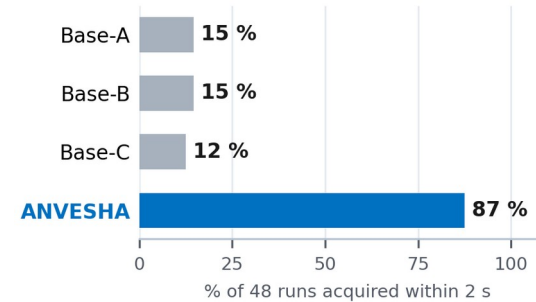
ANVESHA · scenario	Acq. ≤2 s	LOS err	Image err	Loss*	Re-acq	Loop FPS
A Clean sky	3/3 · 1.3 s	1.5	1.5	0 %	—	167
B Gaussian $\sigma=20$	3/3 · 1.2 s	1.9	1.9	0 %	—	166
C Poisson	3/3 · 1.1 s	0.8	0.8	0 %	—	55
D Salt & pepper 10 %	3/3 · 1.1 s	6.7	6.7	0 %	—	142
E Jitter ± 10 px	3/3 · 1.3 s	4.6	8.9	0 %	—	170
F Platform linear	3/3 · 1.1 s	4.4	4.4	0 %	—	167
F2 Vibration 7/13 Hz	2/3 · 1.4 s	20.5	20.5	0 %	—	168
G Fog	3/3 · 1.3 s	1.5	1.5	0 %	—	136
H Haze	3/3 · 1.2 s	3.5	3.5	0 %	—	136
I Low light	3/3 · 1.2 s	2.2	2.2	0 %	—	54
J Fast manoeuvre	3/3 · 1.3 s	18.6	18.6	0 %	—	168
K Occlusion	3/3 · 1.3 s	16.2	16.2	1 %	0.10 s	147
L Rain	3/3 · 1.4 s	1.9	1.9	0 %	—	124
M Combined	3/3 · 1.2 s	4.9	7.8	0 %	—	122
N All PS maxima	0/3	21.0	23.6	99 %	n/a (1X)	51
O Look-alike distractors	1/3 · 2.9 s	1.5	1.5	0 %	—	170

All PS checks met: **ANVESHA 11/16 scenarios** · baselines A/B/C 0/0/0

Limits: acq ≤ 2 s · error ≤ 10 px · loss < 5 % · re-acq ≤ 1 s · ≥ 20 FPS. *excl. scripted occlusion · LOS = excl. camera jitter

Measured gaps — shown, not hidden

- **All PS maxima at once (N):** acquired in 33 % of runs; look-alikes (O) acquired in 2.9 s (> 2 s) — rejecting static look-alikes costs time.
- **LOS error 16–21 px** (> 10) under vibration, fast manoeuvre, occlusion.
- **Jitter floor: ± 20 px** white jitter alone = 16.3 px RMS (analytic) — no coarse gimbal can meet 10 px image error. At ± 10 px (E): image 8.9 vs LOS 4.6 px.

Same seeds, all pipelines**Ablation — remove one part (3 seeds):**

- **belief** → **spiral search**: runs ≤ 2 s 83 % → 7 %
- **adaptive FOV off**: runs ≤ 2 s 83 % → 20 %
- **jitter-aware R off (E)**: LOS 4.6 → 19.3 px
- **IMU feed-fwd off (F)**: LOS 4.4 → 5.8 px
- **CNN verifier off**: false-lock frames 614 → 667 (sum)

Re-acquisition (K, 2 occlusions): mean 0.10 s, max 0.27 s, 0 of 6 unrecovered

Benchmark-2 video rehearsal: centroid RMSE 0.13 px over 300 frames of 2000×2000 @30 fps; 88 FPS incl. decoding

Median acquisition when acquired: **ANVESHA 1.3 s** · baselines 6.4–9.2 s

Potential challenges and risks	Strategies for overcoming these challenges
Sim-to-real gap	video mode already accepts real footage; tracker needs only frames, encoder / IMU and rate commands → HIL
± 20 px jitter floor	report LOS vs image error; jitter-aware R; fine stage (FSM) hand-off
Faint beacon, clutter, look-alikes	joint position × detectability belief; CNN veto + shape tests; static look-alike rejection; beacon modulation ID (next)
CPU budget	ROI processing; CNN only on candidate patches; FPS logged each frame
CNN domain shift	CNN only verifies; classical MF-CFAR stays primary

Implementation status**Implemented**

sim, 16 scenarios, 4 pipelines, belief search, IMM, GUI, video mode, reports

Prototype

CNN verifier: recall 90 %, FPR 23 % on held-out synthetic patches

Planned

.exe build · HIL rig · real footage

❖ Potential impact on the target audience

- 1

ISRO / DoS PAT teams: design & regression-test coarse PAT for mobile ground terminals, UAV/HAP relays and LEO links before hardware exists
- 2

Academia & students: seeded, repeatable PAT testbed — no optics lab, camera or gimbal needed
- 3

Indian FSOC start-ups: tracker talks only frames, encoder/IMU and rate commands → swap simulator for a real camera + pan-tilt (HIL)

❖ Benefits of the solution (social, economic, environmental, etc.)

- **Economic** — develop and tune PAT logic without cameras, pan-tilt units, beacons or an optical bench — the cost barrier named in the PS
- **Technical** — disturbance sweeps that cannot be reproduced outdoors; every run = (scenario, pipeline, seed) with truth-based logs
- **Strategic** — indigenous, auditable coarse-PAT software for Indian optical-communication missions
- **Social / skills** — hands-on PAT, computer vision and control for students across India
- **Environmental** — fewer vehicle / field trials for early tuning

What makes ANVESHA different

Capability	Typical public approach*	ANVESHA
Probability-map (belief) search instead of a fixed spiral	✗	✓
FOV chosen from measured noise & beacon detectability	✗	✓
Re-acquisition seeded from the tracker's prediction	✗	✓
Camera jitter separated from correctable LOS error	✗	✓
Platform IMU feed-forward	some	✓
Same-seed baselines + ablation in one harness	✗	✓
Benchmark-2 video (.mp4) bypass mode	✓	✓
GUI + automatic performance report	✓	✓

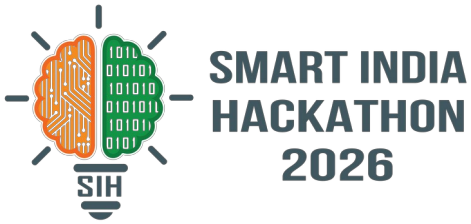
**majority pattern in the READMEs of 20 public SIH26169 repositories · no team named*

Why acquisition matters: an on-orbit optical link reported a 22 s acquisition time [Wang et al., Opt. Lett. 2023].





RESEARCH AND REFERENCES



❖ Details / Links of the reference and research work

134

papers in a structured literature database (DOIs)

20

public SIH26169 repositories analysed for differentiation

16×4×3

scenarios × pipelines × seeds + 7-way ablation

[1] Kaymak Y. et al., “A Survey on Acquisition, Tracking, and Pointing Mechanisms for Mobile FSO Communications,” IEEE COMST, 2018. doi.org/10.1109/comst.2018.2804323

[2] Abdelfatah R. et al., “A review on PAT approaches in UAV-based FSO communication systems,” Opt. Quantum Electron., 2022. doi.org/10.1007/s11082-022-03968-2

[3] Wang X. et al., “On-orbit space optical communication demonstration with a 22 s acquisition time,” Opt. Lett., 2023. doi.org/10.1364/ol.505966

[4] Friederichs L. et al., “Vibration Influence on Hit Probability During Beaconless Spatial Acquisition,” J. Lightwave Technol., 2016. doi.org/10.1109/jlt.2016.2542918

[5] Wang T. et al., “Probability Descent-Based Equivalent Spot Scanning for Satellite Laser Acquisition...,” IEEE TAES, 2025. doi.org/10.1109/taes.2025.3596959

[6] Skryja P., Barcik P., “Optimal Scanning Pattern for Initial Free-Space Optical-Link Alignment,” Photonics, 2024. doi.org/10.3390/photonics11060540

[7] Wang Q. et al., “Approach for recognizing and tracking beacon in inter-satellite optical communication...,” Opt. Express, 2018. doi.org/10.1364/oe.26.028080

[8] Lim H. et al., “Centroid Error Analysis of Beacon Tracking under Atmospheric Turbulence...,” Remote Sensing, 2021. doi.org/10.3390/rs13101931

[9] Li L. et al., “Noise adaptive fading Kalman filter for free-space laser communication beacon tracking,” Appl. Opt., 2016. doi.org/10.1364/ao.55.008486

[10] Youn W., Myung H., “Robust Interacting Multiple Model With Modeling Uncertainties for Maneuvering Target Tracking,” IEEE Access, 2019. doi.org/10.1109/access.2019.2915506

[11] Bian Q. et al., “Accelerometer-Assisted Disturbance Feedforward Control of an Inertially Stabilized Platform...,” IEEE Sensors J., 2023. doi.org/10.1109/jsen.2023.3259446

[12] Kiesbye J. et al., “Hardware-In-The-Loop and Software-In-The-Loop Testing of the MOVE-II CubeSat,” Aerospace, 2019. doi.org/10.3390/aerospace6120130

SOLUTIONS ALREADY EXIST

- Open-loop spiral / raster / Lissajous scans [5][6]
- Wide-FOV acquisition switched by fixed rules
- Threshold/blob + Kalman + PID trackers (most public repos)
- **Blind ~8–10 s search vs PS ≤ 2 s**

➔

ANVESHA STANDS OUT

- **One belief:** search → track → loss → re-acquire
- **Adaptive FOV** from measured noise + beacon detectability
- **Measured:** all PS checks met in 11/16 scenarios; baselines 0/16

Based on our review of 134 papers and 20 public SIH26169 repositories, we did not identify a coarse-PAT system that combines a recursively updated belief map, FOV selection from measured noise, and belief-seeded re-acquisition.

PROJECT RESOURCES

- **Prototype:** run_gui.bat (offline, laptop)
- **Benchmark:** results/bench/benchmark_report.html
- **Source code:** [GitHub link](#) — add before upload
- **Demo video:** [link](#) — add before upload